

Comparison Geometry of SOF Reports

Alignment, Fixed-Fiber Pseudometrics, and Word/Lie Transport Separation

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This paper is Part XIII of the RIME program. Paper VIII defines the static Sectorized Observable Framework (SOF), Paper IX studies observable deformations, Paper X introduces the Universal Observable Pipeline and the SOF Registry, Paper XI classifies observable walls, and Paper XII formulates the SOF Diagnostic Protocol for single-system SOF Reports. Paper XIII introduces the local comparison geometry of SOF reports. It asks why comparison is impossible before alignment, what geometry exists on a fixed fiber of aligned structural representations, and which observable transport channels must remain distinct.

Abstract

Problem. Paper XII defines a single-system SOF Report, but two reports are not comparable merely because they conform to the same reporting protocol. Statements such as “support decreased” remain undefined until sectors, observables, thresholds, depth conventions, and parameter samples have been aligned. Paper XIII asks what mathematical comparison object exists after these choices are made explicit.

Approach. The alignment datum is

$$\mathfrak{A}_{\text{align}} = (\mathcal{R}^*, \widehat{\mathcal{R}}, \Phi_{\text{sec}}, \Phi_{\text{obs}}).$$

Here $\mathcal{R}^*$ and $\widehat{\mathcal{R}}$ are the reference and target reports and $\Phi_{\text{sec}}, \Phi_{\text{obs}}$ align their sectors and observables in common retained coordinates. The canonical SOF comparison object is

$$\mathfrak{C}_{\text{cmp}} = (\mathfrak{A}_{\text{align}}, \Theta) = (\mathcal{R}^*, \widehat{\mathcal{R}}, \Phi_{\text{sec}}, \Phi_{\text{obs}}; \Theta),$$

where Θ fixes normalization, thresholds, depth semantics, path synchronization, metrics, and aggregation. Comparison then induces

$$\Delta_{\text{audit}} = \text{Compare}(\mathfrak{C}_{\text{cmp}}) = \text{Compare}_{\Theta}(\mathfrak{A}_{\text{align}}).$$

The report pair alone is not comparable, the alignment datum alone does not fix comparison semantics, and the audit signature is an output rather than the comparison object. The machine-readable record retains $\mathfrak{C}_{\text{cmp}}$ and Δ_{audit} separately.

Results. On each fixed fiber of aligned structural representations, the structural signatures carry weighted Hamming pseudometrics; strictly positive weights give a metric after quotienting by equality of the retained structural signature. This is a local comparison geometry, not a metric on the full eight-coordinate audit signature. An exact-support lemma places Lie-bridge support inside the union of the two ordered length-two word supports, while a single-generator X^2 witness proves that word accessibility can exist with no Lie accessibility. Computational validation comprises 25 constructed failure-mode comparisons across GridWorld, SIR, Traffic, Compiler IR, and the additional Network Routing domain; the four main domains distinguish their five declared controls by signature pattern. In the GridWorld F4 control, the Lie channel changes on 8 ordered pairs while the word channel remains unchanged. Three legitimate before/after transformations have nonzero raw signatures but zero residual contract violations.

Implications. Paper XIII introduces the local comparison geometry of SOF reports. Comparison is impossible before alignment; after alignment, report differences become structured coordinates rather than aggregate scores. A nonzero signature records change, not defect. Latent-model and black-box comparison remain deployment regimes rather than evidence claims, and cross-fiber transport remains open.

1 Introduction

Papers VIII–XI establish the SOF object language, observable dynamics, Universal Observable Pipeline, Registry, and wall-record taxonomy [4, 3, 6, 2]. Paper XII then turns that language into a single-system reporting protocol: a declared sectorization and observable family produce a claim-status-aware `.sofreport` record [5].

Paper XIII changes the question from reporting to comparison. A single report answers: *given this sectorization and these observables, what structure was observed?* It cannot answer whether that structure agrees with a reference. Correctness, conformance, and change are relational notions; two report files are not yet a mathematical comparison object.

Before alignment, even a basic statement such as “the target has less support” has no invariant meaning. The apparent difference may arise because:

- the two reports use different sectors;
- their observable families are not in correspondence;
- their thresholds or depth semantics differ;
- their parameter samples or trajectory samples are not synchronized.

The pair $(\mathcal{R}^*, \widehat{\mathcal{R}})$ is therefore insufficient. The alignment datum

$$\mathfrak{A}_{\text{align}} = (\mathcal{R}^*, \widehat{\mathcal{R}}, \Phi_{\text{sec}}, \Phi_{\text{obs}})$$

places the reports in common retained coordinates. The additional specification Θ fixes how those coordinates are compared. Together they form the SOF comparison object

$$\mathfrak{C}_{\text{cmp}} = (\mathfrak{A}_{\text{align}}, \Theta).$$

SOF Report Alignment Principle. Compare a target report $\widehat{\mathcal{R}}$ against a reference report $\mathcal{R}^*$ only after sector and observable alignments Φ_{sec} and Φ_{obs} are explicit. The admissible comparison object is $(\mathfrak{A}_{\text{align}}, \Theta)$; Δ_{audit} is its induced comparison signature.

Paper XIII makes four contributions:

1. It defines the SOF Report Alignment datum and the canonical comparison object $\mathfrak{C}_{\text{cmp}} = (\mathfrak{A}_{\text{align}}, \Theta)$, making comparability explicit rather than hidden preprocessing.
2. It defines a fiber of aligned structural representations and equips it with a weighted structural pseudometric, giving a local comparison geometry.
3. It proves exact word/Lie support inclusion and a strict word-only witness, establishing that combinatorial and algebraic transport require separate signature channels.

From SOF Reports to an Audit Signature

Alignment and comparison semantics form the canonical comparison object

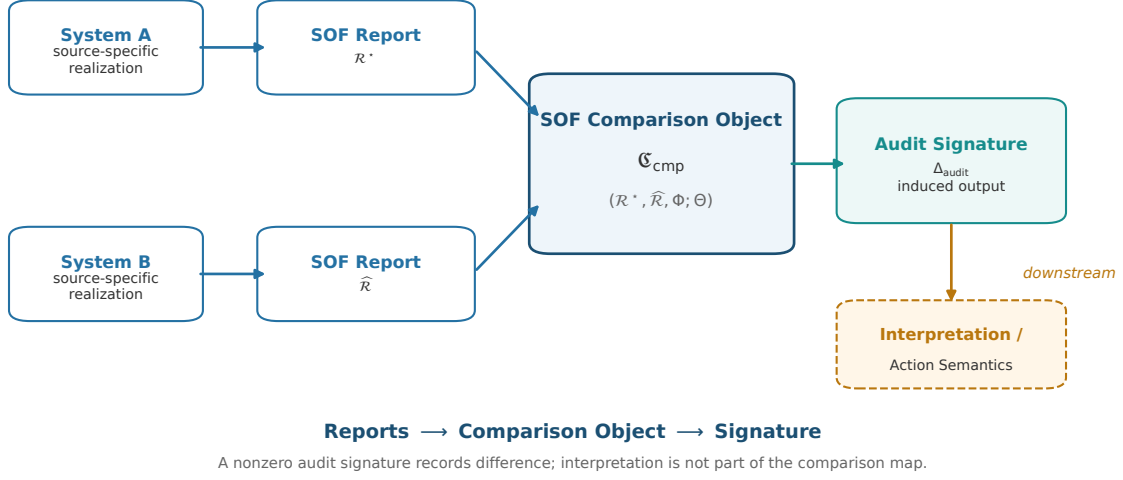


Figure 1. From SOF Reports to an audit signature. Two independently generated SOF Reports, their explicit alignment, and the comparison specification form $\mathfrak{C}_{\text{cmp}}$. The induced Δ_{audit} records difference; interpretation and Action Semantics remain downstream.

4. It validates one comparison grammar across GridWorld, SIR, Traffic, Compiler IR, and an additional Network Routing control, including legitimate transformations whose nonzero differences are licensed by contract.

The theory developed here is local. The paper does not define comparison between different alignment fibers, infer alignments automatically, prove a metric on the full audit signature, or turn difference into defect or action. Single-report production belongs to Paper XII, aligned comparison to Paper XIII, and context-indexed interpretation and action semantics to Paper XIV.

The paper proceeds from the comparison object to the induced audit signature, its fixed-fiber structural geometry, and word/Lie separation. It then presents the controlled validations, portability analysis, and failure boundaries. The appendices state the machine-readable alignment contract, reproducibility map, and geometric boundary; broader alignment regimes are deferred to the Outlook.

2 SOF Report Alignment

2.1 Alignment Datum and Comparison Object

For a report $\mathcal{R}$, write $I(\mathcal{R})$ for its retained sector-label set and $G(\mathcal{R})$ for its retained observable-label set. An alignment places the two reports in common retained coordinate sets I_Φ and G_Φ . Schematically, its sector component is the span

$$I(\mathcal{R}^*)_{\text{ret}} \xrightarrow{\phi_{\text{sec}}^*} I_\Phi \xleftarrow{\hat{\phi}_{\text{sec}}} I(\widehat{\mathcal{R}})_{\text{ret}},$$

and its observable component is defined analogously over G_Φ . The legs may be identities, relabellings, or declared aggregations. They need not be total or bijective, but unmatched coordinates and any aggregation rule must be explicit.

The SOF Report Alignment datum is

$$\mathfrak{A}_{\text{align}} = (\mathcal{R}^*, \widehat{\mathcal{R}}, \Phi_{\text{sec}}, \Phi_{\text{obs}}), \quad \Phi = (\Phi_{\text{sec}}, \Phi_{\text{obs}}).$$

Definition 2.1 (SOF Report Alignment).

A SOF Report Alignment consists of a reference report, a target report, a sector alignment, and an observable alignment into common retained coordinates. It is the minimal geometric datum required before a domain-independent report comparison can be specified.

The comparison specification is written schematically as

$$\Theta = (\mathbf{N}, \mathbf{M}, \mathbf{S}_D, \tau, \mathbf{P}, \mathbf{A}),$$

where $\mathbf{N}$ specifies normalization and scaling, $\mathbf{M}$ specifies the comparison metric, $\mathbf{S}_D$ fixes depth semantics and frozen-value handling, τ records thresholds and tolerances, $\mathbf{P}$ synchronizes parameterized paths, and $\mathbf{A}$ specifies aggregation. Components fixed by a schema version or protocol default need not be repeated in every record, but they remain part of the declared comparison semantics.

Definition 2.2 (SOF comparison object).

A SOF comparison object is

$$\mathfrak{C}_{\text{cmp}} = (\mathfrak{A}_{\text{align}}, \Theta) = (\mathcal{R}^*, \widehat{\mathcal{R}}, \Phi_{\text{sec}}, \Phi_{\text{obs}}; \Theta).$$

It is **admissible** when the report references resolve, Φ induces shape-compatible common coordinates for every compared structural array, unmatched coordinates are declared, and Θ fixes the semantics of every non-null comparison coordinate. Path-dependent coordinates additionally require declared parameter synchronization.

Comparison is the operator

$$\text{Compare} : \text{SOFComparison} \longrightarrow \text{AuditSignature},$$

which maps an admissible comparison object to its comparison signature:

$$\boxed{\Delta_{\text{audit}} = \text{Compare}(\mathfrak{C}_{\text{cmp}}) = \text{Compare}_{\Theta}(\mathfrak{A}_{\text{align}})}.$$

The second equality is the curried notation used when Θ is fixed. Thus Δ_{audit} is induced rather than primitive. Its serialized record retains the comparison object and resulting signature as distinct fields.

Remark 2.3 (Comparison-object non-uniqueness).

Alignment is not unique in general. Different admissible choices of Φ_{sec} or Φ_{obs} may produce different comparison signatures under the same Θ . Different admissible choices of Θ may also change the signature under the same alignment. Both Φ and Θ therefore remain inside the `.sofaudit` comparison object rather than hidden preprocessing.

2.2 The Alignment Contract

Without Φ_{sec} and Φ_{obs} , two reports are merely two unrelated descriptions. The alignment contract establishes comparability:

- **Sector alignment** Φ_{sec} : In the simplest case (same ambient space, same sector partition), this is the identity. When sector counts differ or sectorizations are produced by different pipelines (e.g., reference from simulation, target from clustering), Φ_{sec} must specify which reference sector each target sector corresponds to, or declare pairs incomparable.
- **Observable alignment** Φ_{obs} : When observable families differ in cardinality or semantics (e.g., reference has 4 directional controls, target has 4 but in a different order), Φ_{obs} specifies the correspondence. When observables are incommensurable (reference: physical forces; target: neural network activations), alignment is defined at the sector-block support level rather than the operator level.

In the controlled validations below, both alignments are identity maps or explicitly declared block/observable correspondences. Latent and behavioral comparisons generally require non-trivial alignments.

2.3 Notation

We distinguish SOF objects from SOF reports:

Symbol	Meaning
$\mathcal{F}$	A SOF object: the triple $(V, \{Q_i\}, \{X_k\})$
$\mathcal{R}$	A SOF Report: the diagnostic record produced from $\mathcal{F}$
$\mathcal{R}^*$	Reference report
$\widehat{\mathcal{R}}$	Target report
$\mathfrak{A}_{\text{align}}$	SOF Report Alignment datum
$\mathfrak{C}_{\text{cmp}}$	Admissible SOF comparison object $(\mathfrak{A}_{\text{align}}, \Theta)$
Θ	Comparison specification: normalization, metric, depth semantics, thresholds, parameter synchronization, and aggregation
$\mathfrak{F}_{\Phi, \Theta}$	Fiber of structural representations expressed under fixed (Φ, Θ)
$\text{Sig}_{\Phi, \Theta}^{\text{str}}(\mathcal{R}^\epsilon)$	Aligned structural representation of one side $\epsilon \in \{\text{ref}, \text{tar}\}$
$d_{\Phi, \Theta}$	Fixed-fiber structural pseudometric
SOFComparison	Class of admissible SOF comparison objects
AuditSignature	Space of structured comparison signatures
Δ_{audit}	Mathematical comparison signature induced by alignment and Θ

This convention separates SOF objects from their report-level realizations. The `.sofaudit` file serializes $\mathfrak{C}_{\text{cmp}}$ together with its induced Δ_{audit} and factual claim boundary.

3 Audit Signature and Fixed-Fiber Geometry

3.1 Comparison Signature

Once Φ_{sec} and Φ_{obs} are fixed, the induced comparison signature is an eight-coordinate structured comparison:

$$\Delta_{\text{audit}} = (\Delta_{\text{supp}}, \Delta_{\text{brw}}, \Delta_{\text{brl}}, \Delta_{\text{dep}}, \Delta_{\text{frz}}, \Delta_{\text{cns}}, \Delta_{\text{ctrl}}, \Delta_{\text{wal}})$$

The Alignment-Induced Audit Signature

Eight coordinate groups retain typed differences instead of collapsing them into one score

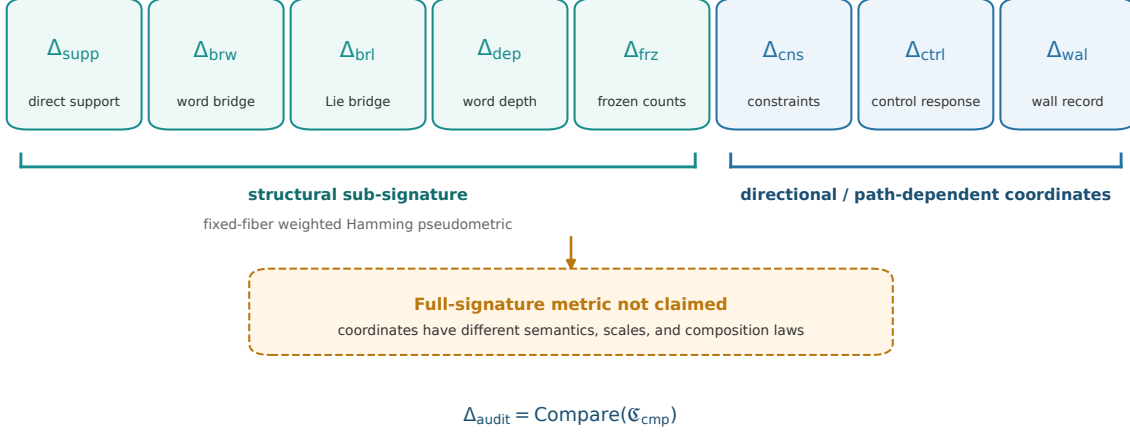


Figure 2. The alignment-induced audit signature. The first five coordinate groups form the structural sub-signature used by the fixed-fiber pseudometric. Constraint, control-response, and wall-record coordinates retain directional or path-dependent semantics. No metric on the full signature is claimed.

Dimension	Symbol	What is compared	What it detects
Support mismatch	Δ_{supp}	R_1^* vs. $\widehat{R}_1$	Missing or hallucinated direct transitions
Bridge mismatch (word)	Δ_{brw}	$R_{2,\text{word}}^*$ vs. $\widehat{R}_{2,\text{word}}$	Lost or spurious 2-step word paths
Bridge mismatch (Lie)	Δ_{brl}	$R_{2,\text{Lie}}^*$ vs. $\widehat{R}_{2,\text{Lie}}$	Commutator-structure changes
Depth distortion	Δ_{dep}	D_{word}^* vs. $\widehat{D}_{\text{word}}$	Path-length errors and pairwise frozen/reachable misclassification
Frozen disagreement	Δ_{frz}	$(f_{R_1}, f_{D,\text{word}}, f_{D,\text{Lie}})^*$ vs. target counts	Net over/under-estimation of direct, word-depth, and Lie-depth reachability (three coordinates)
Constraint violations	Δ_{cns}	Raw T^* vs. target T	Target transitions absent from the reference
Control-response mismatch	Δ_{ctrl}	Per-control $c_a^*(i, j)$ vs. $\widehat{c}_a(i, j)$	Rate errors, native-control aliasing, response collapse
Wall-record mismatch	Δ_{wal}	Wall records along parameterized path	Failure to track environmental deformation

3.2 Why Eight, Not Seven

The comparison signature has eight coordinate groups. Δ_{frz} is one group with three coordinates (frozen_R1, frozen_D_word, and frozen_D_lie), and Δ_{wal} is the eighth group. The wall record

is a distinct diagnostic object: a parameterized family of signatures rather than a single-pair comparison.

The displayed coordinate Δ_{ctrl} concerns native system controls such as GridWorld moves, epidemiological rates, or traffic phases. It does not denote a downstream intervention action. The v1.0 machine-readable contract retains the field name `action_response_failure` as a compatibility alias for this control-response coordinate.

3.3 Aligned Structural Representation Fiber

Fix a comparison frame (Φ, Θ) , including the common retained sector and observable index sets induced by Φ . For either side $\epsilon \in \{\text{ref}, \text{tar}\}$ of an admissible comparison object, with $\mathcal{R}^{\text{ref}} = \mathcal{R}^*$ and $\mathcal{R}^{\text{tar}} = \widehat{\mathcal{R}}$, let

$$\text{Sig}_{\Phi, \Theta}^{\text{str}}(\mathcal{R}^\epsilon) = (R_1, R_{2, \text{word}}, R_{2, \text{Lie}}, D_{\text{word}}, f_{R_1}, f_{D, \text{word}}, f_{D, \text{Lie}}).$$

Here every matrix has first been reindexed or aggregated into the common coordinates declared by Φ according to Θ . The first four coordinates are aligned matrices. The final three are the canonical frozen coordinates retained by Δ_{fz} . Write $\text{Adm}(\Phi, \Theta)$ for the report sides admissibly embedded in this fixed comparison frame.

Definition 3.1 (Aligned structural representation fiber).

The fixed fiber $\mathfrak{F}_{\Phi, \Theta}$ is the collection of aligned structural representations

$$\mathfrak{F}_{\Phi, \Theta} = \left\{ \text{Sig}_{\Phi, \Theta}^{\text{str}}(\mathcal{R}^\epsilon) : \mathcal{R}^\epsilon \in \text{Adm}(\Phi, \Theta), \quad \epsilon \in \{\text{ref}, \text{tar}\} \right\}.$$

Its elements are aligned structural representations, not unaligned SOF Reports and not audit deltas. Representations expressed under different alignment or comparison specifications belong to different fibers.

For a matrix coordinate c , let H_c be off-diagonal entrywise Hamming distance; for a frozen-count coordinate, let H_c be the discrete distance. Given weights $w_c \geq 0$ and $\sigma, \sigma' \in \mathfrak{F}_{\Phi, \Theta}$, set

$$d_{\Phi, \Theta}(\sigma, \sigma') = \sum_{c \in \mathcal{C}_{\text{str}}} w_c H_c(\sigma_c, \sigma'_c).$$

Proposition 3.2 (Fixed-fiber structural pseudometric).

The function

$$d_{\Phi, \Theta} : \mathfrak{F}_{\Phi, \Theta} \times \mathfrak{F}_{\Phi, \Theta} \longrightarrow \mathbb{R}_{\geq 0}$$

is a pseudometric. If every $w_c > 0$, it is a metric on the retained aligned structural signature tuples, or equivalently on richer aligned representations quotiented by equality of those tuples. Omitted diagonal entries are part of the declared equivalence.

Proof. Every H_c is nonnegative and symmetric. Pointwise mismatch obeys

$$\mathbf{1}[x \neq z] \leq \mathbf{1}[x \neq y] + \mathbf{1}[y \neq z],$$

so each matrix Hamming coordinate satisfies the triangle inequality; the discrete frozen-count coordinates do as well. A nonnegative weighted sum preserves these properties. Zero-weight coordinates may identify distinct signatures, giving a pseudometric. If all weights are positive, zero distance is equivalent to equality of every retained structural coordinate.

This proposition applies only to the **structural sub-signature**. It does not establish a metric on the full eight-coordinate Δ_{audit} : constraint violations and control-response mismatches may be reference-to-target directional, while wall comparison requires path synchronization. Those coordinates require separate symmetrization, normalization, and composition laws before a full-signature geometry can be claimed.

The fixed-fiber theory is local. Each $\mathfrak{F}_{\Phi, \Theta}$ is a local comparison domain in which coordinates and mismatch semantics have already been fixed. No transition law between distinct fibers is defined; cross-fiber geometry requires compatible pushforwards and coordinate-change laws.

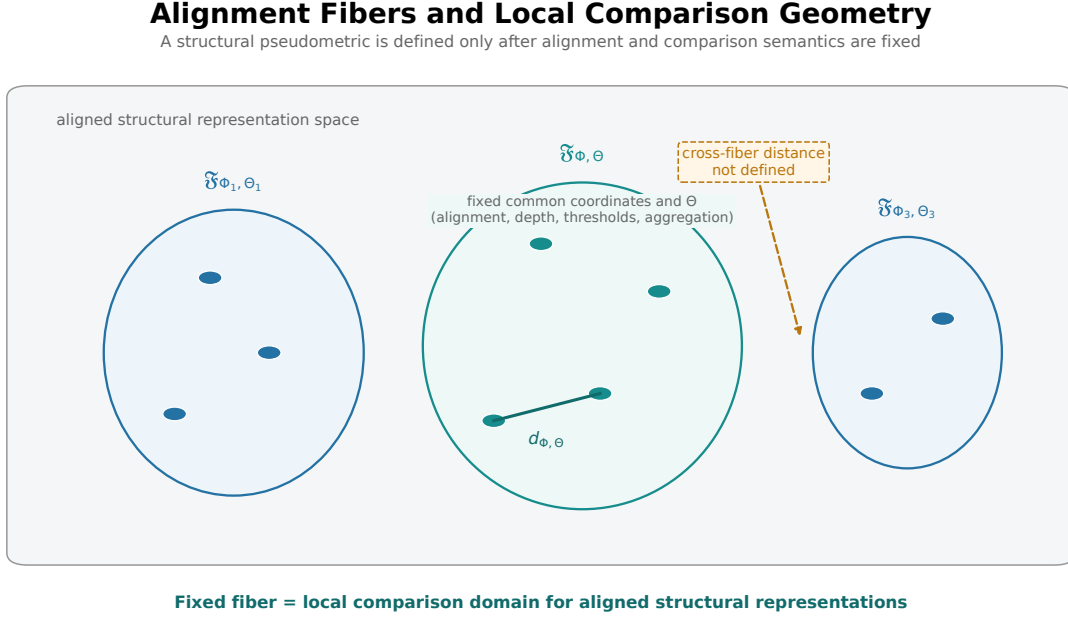


Figure 3. Alignment fibers and local comparison geometry. Each fiber fixes sector and observable alignments together with the comparison semantics in Θ . The weighted structural pseudometric is defined within a local comparison domain; the dashed cross-fiber relation is intentionally left undefined.

3.4 Word Bridge vs. Lie Bridge

Word bridges and Lie bridges capture different algebraic transport mechanisms. The word channel is generated by products such as $X_a X_b$, while the Lie channel is generated by commutators such as $[X_a, X_b]$. Their responses to the same perturbation need not coincide, so neither channel should be inferred from the other. They are therefore reported as separate coordinates of AuditSignature.

For the declared sectorization, write

$$\text{supp}_Q(Y) = \{(i, j) : Q_i Y Q_j \neq 0, i \neq j\}.$$

Lemma 3.3 (Exact word/Lie support inclusion).

For every pair of observables,

$$\text{supp}_Q([X_a, X_b]) \subseteq \text{supp}_Q(X_a X_b) \cup \text{supp}_Q(X_b X_a).$$

Proof. If both $Q_i X_a X_b Q_j$ and $Q_i X_b X_a Q_j$ vanish, then their difference $Q_i [X_a, X_b] Q_j$ vanishes. Taking the contrapositive proves the inclusion.

The lemma is an **exact-support** statement. At finite absolute threshold τ , one has only

$$\|Q_i [X_a, X_b] Q_j\|_F \leq \|Q_i X_a X_b Q_j\|_F + \|Q_i X_b X_a Q_j\|_F.$$

Consequently, Lie support detected at threshold τ guarantees that at least one ordered word block exceeds $\tau/2$, not necessarily τ . Thresholded word/Lie comparisons therefore require compatible thresholds inside Θ .

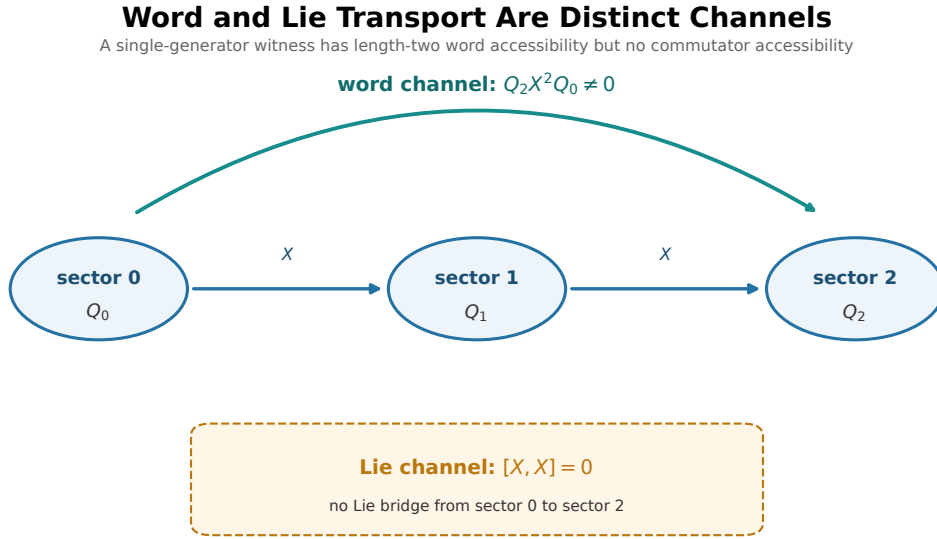
The inclusion is strict. Let three one-dimensional sectors be connected by the single skew-Hermitian generator

$$X = E_{10} - E_{01} + E_{21} - E_{12}.$$

Then

$$Q_2 X^2 Q_0 = E_{20} \neq 0, \quad [X, X] = 0.$$

Thus a length-two word channel connects sector 0 to sector 2 while the Lie channel is absent. Word-depth accessibility can therefore exist without Lie-channel accessibility, so the two channels cannot be merged into one depth coordinate.



Combinatorial composition and algebraic commutation cannot be merged into one depth channel.

Figure 4. Strict word/Lie separation witness. The single generator X connects successive sectors, so $Q_2 X^2 Q_0 \neq 0$, while the only self-commutator $[X, X]$ vanishes. Word accessibility and Lie accessibility are therefore distinct transport channels.

This separation is species-independent. It distinguishes two registry-level transport mechanisms:

$$\begin{aligned} \text{combinatorial transport} &\leftrightarrow \text{word composition,} \\ \text{algebraic transport} &\leftrightarrow \text{Lie commutators.} \end{aligned}$$

They may coexist in one realization, but neither is a surrogate for the other. The distinction belongs to the SOF observable language rather than to GridWorld, Compiler IR, or any other validation domain.

The controlled F4 example demonstrates this distinction. On the tested skew-symmetrized generators, word products retain connectivity through reverse-direction components while the commutator channel changes. Hence $\Delta_{\text{brw}} = 0$ and $\Delta_{\text{brl}} > 0$ can occur in the same comparison. This is evidence for separate reporting, not a universal sensitivity ordering between word and Lie bridges.

3.5 Machine-Readable Comparison Record

Paired comparisons are serialized as `.sofaudit` records, distinct from the single-system `.sofreport` format of Paper XII. A `.sofaudit` record contains the canonical comparison object $\mathfrak{C}_{\text{cmp}} = (\mathcal{R}^*, \mathcal{R}, \Phi; \Theta)$, the induced signature Δ_{audit} , and its factual claim boundary. It is a companion comparison contract rather than a revision of SOFRS. The serialized field `comparison_object` identifies the input object; `signature` identifies the output. Appendix A states the contract semantics; the canonical JSON Schema is maintained separately.

3.6 Downstream Interpretation Boundary

The comparison signature and its downstream interpretation are different objects. For a declared context Γ , a downstream semantic layer may interpret each coordinate through

$$\text{Sem}_{\Gamma,i} : \Delta_i \longrightarrow \mathcal{J}_i.$$

Here $\mathcal{J}_i$ contains a context-relative interpretation label and may carry declared severity or confidence annotations. It contains no repair command, intervention candidate, or policy selection. Paper XIII records only the factual alignment and signature; it does not define $\text{Sem}_{\Gamma,i}$ inside the comparison geometry.

Paper XIV owns the semantic and action layers:

$$\Delta_i \xrightarrow{\text{Sem}_{\Gamma,i}} \mathcal{J}_i, \quad (\Delta_{\text{audit}}, \text{Sem}_{\Gamma}) \mapsto \mathcal{A}_{\Gamma}(\Delta_{\text{audit}}).$$

Any subsequent policy selection acts on the candidate Action Set and is further downstream. Neither interpretation nor action may overwrite the recorded Δ_{audit} .

3.7 Legitimate Transformation Contracts

A nonzero comparison signature is descriptive, not condemnatory. For a declared legitimate transformation, the paired comparison may additionally record a transformation contract

$$\mathcal{T} = (\text{intent}, \text{allowed changes}, \text{preserved invariants}, \text{required postconditions}).$$

The raw Δ_{audit} remains unchanged. A downstream contract evaluation records only residual changes not licensed by $\mathcal{T}$. Thus a large structural signature may still be conforming, while a small signature may violate a required invariant. The `transformation_contract` and `contract_evaluation` fields are optional `.sofaudit` metadata and do not alter the intrinsic coordinates of Δ_{audit} .

4 Controlled Validation: GridWorld

4.1 GridWorld Setup

The reference is a deterministic 5×5 grid with 25 one-hot cell sectors and an absorbing obstacle at (2, 2). Moves originating at the obstacle self-loop, while neighboring moves into it are blocked.

The observable family consists of the four skew-symmetrized directional transition matrices $\{N, S, E, W\}$. Five target perturbations test distinct comparison channels.

4.2 GridWorld Failure Modes

F1 – Action Aliasing ($N = S$). The target model’s S transition matrix is replaced by N . Because the response constants retain block norms rather than orientation or sign, the support, bridge, depth, frozen, and response channels do not detect this opposite-action aliasing; only the raw transition-constraint channel responds. This exposes a boundary of the chosen observable family.

F2 – Persistence Loss. E from $(1,1)$ is smeared across three targets $(0.5, 0.3, 0.2)$ weights). The new edges lie on already-supported sector pairs, so direct and word support remain fixed while commutator paths change.

F3 – Forbidden Edge. S from $(1,2)$ hallucinates a transition into the absorbing obstacle sector $(2,2)$. This adds direct support, creates word and Lie bridges, shortens word depth, and violates the reference transition constraint.

F4 – Rare Bridge Deletion. E removed from all column-0 cells $(0, 5, 10, 15, 20)$. Reverse-oriented skew components preserve the registered word paths, while the commutator channel and Lie-depth shadow change.

F5 – Obstacle Deformation. Reference sweeps obstacle position $(2,2) \rightarrow (2,1) \rightarrow (2,0) \rightarrow (1,0) \rightarrow (0,0)$. The target remains fixed at $\text{obstacle}=(2,1)$. The path records support, bridge, depth, and frozen-set changes; equal net frozen counts at some steps still conceal different frozen pairs.

4.3 GridWorld Signature Table

Failure	Δ_{supp}	Δ_{brw}	Δ_{brl}	Δ_{dep}	$\Delta_{\text{frz}} = (R_1, W, L)$	Δ_{cns}	Δ_{ctrl}	Δ_{wal}
F1 action aliasing	0	0	0	0	$(0, 0, 0)$	18	0	–
F2 persistence loss	0	0	18	0	$(0, 0, -78)$	2	6	–
F3 forbidden edge	2	6	6	48	$(-2, -48, -66)$	1	2	–
F4 bridge deletion	0	0	8	0	$(0, 0, -52)$	0	10	–
F5 deformation	12	30	16	108	$(0, 0, +20)$	6	24	5 steps

The five controls have distinct coordinate-activation patterns. In particular, F4 has $\Delta_{\text{brw}} = 0$ and $\Delta_{\text{brl}} = 8$, giving direct evidence that word and Lie bridge channels must be reported separately. This control does not imply a universal sensitivity ordering.

5 Cross-Domain Controlled Validation

Each domain supplies its own sectorization, observable family, and alignment contract while retaining the same comparison object and signature grammar. No equivalence of the native dynamics or failure mechanisms is assumed.

5.1 SIR Compartmental Model

5.1.1 SIR Realization

The reference SIR system has three one-hot compartment sectors, Susceptible, Infectious, and Recovered, with rate observables β for $S \rightarrow I$ and γ for $I \rightarrow R$. The reference parameters are $\beta = 0.3$ and $\gamma = 0.1$, so $R_0 = 3$. The directed chain is $S \xrightarrow{\beta} I \xrightarrow{\gamma} R$. Under the skew-symmetrized rate generators:

- R_1 : $S \leftrightarrow I$ and $I \leftrightarrow R$ each contribute 2 ordered off-diagonal pairs, yielding $R_1^{\text{offdiag}} = 4$. The skew part records antisymmetric support: $X_\beta[I, S] = \beta/2$ and $X_\beta[S, I] = -\beta/2$ together encode the S – I support channel. This is a proxy for the directed epidemiological $S \rightarrow I$ semantic, not a direct encoding.
- $R_{2,\text{word}}$: $S \leftrightarrow R$ via the 2-step word bridge $X_\gamma X_\beta$. $R_{2,\text{word}}^{\text{offdiag}} = 2$.
- D_{word} : $D(S, I) = 1$, $D(I, R) = 1$, and $D(S, R) = 2$, with symmetric reverse entries.
- $f_{R_1} = 2$ for the directly frozen S – R ordered pairs, while $f_{D,\text{word}} = f_{D,\text{Lie}} = 0$.
- $R_{2,\text{Lie}}$: $[X_\beta, X_\gamma]$ has a non-zero S – R block. $R_{2,\text{Lie}}^{\text{offdiag}} = 2$.

This is a minimal three-sector realization with a nontrivial two-step bridge.

5.1.2 SIR Failure Modes

F1 – Rate Equalization ($\beta = \gamma$). Binary structure is unchanged and only Δ_{ctrl} responds. The infection and recovery operators remain distinct because they occupy different support blocks.

F2 – Missing Edge ($\beta = 0$). The $S \rightarrow I$ channel vanishes, changing direct support, both bridge channels, depth, and all three frozen coordinates.

F3 – Forbidden Direct ($S \rightarrow R$). A spurious direct edge is added to the β observable, producing extra support and bridge paths together with one reference-constraint violation.

F4 – Rate Distortion ($\beta = 0.001$). Structure is preserved ($\beta > 0$ so edges exist), but the $S \rightarrow I$ response is 300 times weaker. Only the continuous control-response coordinate changes.

F5 – Wall Record. β is swept from 0 to 0.5 in 11 steps; the target is fixed at $\beta = 0.2$. The frozen-set mismatch occurs only at the endpoint $\beta = 0$; for all sampled $\beta > 0$, the reference and target have identical frozen sets.

5.1.3 SIR Signature Table

Failure	Δ_{supp}	Δ_{brw}	Δ_{brl}	Δ_{dep}	$\Delta_{\text{frz}} = (R_1, W, L)$	Δ_{cns}	Δ_{ctrl}	Δ_{wal}
F1 rate equalization	0	0	0	0	(0, 0, 0)	0	4	–
F2 missing edge	2	2	2	4	(+2, +4, +4)	0	2	–
F3 forbidden direct	2	4	2	2	(-2, 0, 0)	1	2	–
F4 rate distortion	0	0	0	0	(0, 0, 0)	0	2	–
F5 wall record	0	0	0	0	path-dependent*	0	2	11 steps

* Along the wall path, the frozen-count delta is $(-2, -4, -4)$ at $\beta = 0$ and $(0, 0, 0)$ for every sampled $\beta > 0$.

5.1.4 Two Walls in SIR Parameter Space

The SIR β -sweep reveals two walls of different SOF types:

- $\beta = 0$ (**topological wall**). The $S \rightarrow I$ edge vanishes. R_1 loses the S – I support pair. The direct and depth-frozen counts jump, producing a structural boundary.
- $R_0 = 1$, i.e., $\beta = \gamma$ (**response-order crossing**). $\|X_\beta\|/\|X_\gamma\|$ crosses 1. No binary metric changes: R_1 , R_2 , and frozen sets are constant for all $\beta > 0$. The crossing is visible only in the continuous response constants.

This distinction separates a support wall from a continuous response-order crossing. In classical epidemiology, $R_0 = 1$ is already the critical epidemic threshold. SOF does not replace that interpretation; it re-expresses the same threshold as a crossing of registered response magnitudes rather than a change in compartmental support.

5.2 Traffic Intersection

The traffic-intersection control adds a third Regime A domain. The reference is a directed 2x2 intersection grid with four one-hot sectors (NW, NE, SW, SE) and two signal-phase observables. Phase A records N-S green movement through southbound edges $NW \rightarrow SW$ and $NE \rightarrow SE$. Phase B records E-W green movement through westbound edges $NE \rightarrow NW$ and $SE \rightarrow SW$. Reverse directions are represented by the skew-symmetric transpose, as in the other finite SOF controls.

The deformation parameter is the green-time ratio $\rho = t_A/t_B$. The reference is $\rho = 1$. Five constructed variants are compared:

- **F1 phase aliasing**. Phase B is replaced by Phase A, losing the horizontal response while making the two signal phases indistinguishable.
- **F2 missing phase**. $\rho = 0$, so Phase A is removed and vertical movement freezes.
- **F3 forbidden diagonal**. Phase A hallucinates an $NW \rightarrow SE$ diagonal edge.
- **F4 timing distortion**. $\rho = 0.01$, so binary support is preserved while control-response magnitudes collapse.
- **F5 wall record**. The reference sweeps ρ from 0.01 to 100 while the target is fixed at $\rho = 2$.

Failure	Δ_{supp}	Δ_{brw}	Δ_{brl}	Δ_{dep}	$\Delta_{\text{frz}} = (R_1, W, L)$	Δ_{cns}	Δ_{ctrl}	Δ_{wal}
F1 phase aliasing	4	4	0	8	(+4, +8, +4)	2	8	–
F2 missing phase	4	4	0	8	(+4, +8, +4)	0	8	–
F3 forbidden diagonal	2	8	4	2	(-2, 0, -4)	1	2	–
F4 timing distortion	0	0	0	0	(0, 0, 0)	0	8	–
F5 wall record	0	0	0	0	(0, 0, 0)	0	8	21 steps

Traffic F5 provides rate-order rather than binary wall evidence. The sampled interval $\rho \in [0.01, 100]$ excludes the limit points $\rho \rightarrow 0$ and $\rho \rightarrow \infty$, so frozen-count deltas remain (0, 0, 0) under the declared tolerance. The record is therefore a trajectory-mismatch path, not a binary wall-crossing observation.

5.3 Compiler IR

The compiler-IR control uses two distinct observable families on five one-hot basic-block sectors:

$$(B_{0,\text{entry}}, B_1, B_2, B_{3,\text{side}}, B_{4,\text{exit}}).$$

The observable family contains:

- X_{cfg} , recording control-flow edges;
- X_{defuse} , recording cross-block def-use chains.

The reference is an O2-like five-block IR where the side block B3 is reachable and the $B3 \rightarrow B4$ data dependence is intact. Alignment is compiler-native: basic-block names, SSA provenance, CFG node correspondence, dominance metadata, or debug/source-span metadata can provide the comparison contract. The experiment uses identity alignment.

Five constructed variants are compared:

- **F1 CFG/def-use aliasing.** X_{defuse} is replaced by X_{cfg} , making data-flow structure indistinguishable from control flow.
- **F2 dead branch loss.** B3 is isolated by removing both CFG and def-use edges.
- **F3 spurious CFG edge.** A hallucinated $B0 \rightarrow B4$ jump bypasses the normal path.
- **F4 lost def-use.** The $B3 \rightarrow B4$ data edge is removed while the CFG edge is preserved.
- **F5 pass-pipeline wall.** The reference follows $O0 \rightarrow \text{mem2reg} \rightarrow \text{simplifycfg}$, while the target remains fixed at the pre-simplifycfg snapshot.

Failure	Δ_{supp}	Δ_{brw}	Δ_{brl}	Δ_{dep}	$\Delta_{\text{frz}} = (R_1, W, L)$	Δ_{cns}	Δ_{ctrl}	Δ_{wal}
F1 CFG/def-use aliasing	2	10	12	4	(+2, 0, +10)	2	6	–
F2 dead branch loss	4	8	8	8	(+4, +8, +8)	0	6	–
F3 spurious CFG edge	2	2	4	2	(-2, 0, 0)	1	2	–
F4 lost def-use	0	0	4	0	(0, 0, 0)	0	2	–
F5 pass-pipeline wall	0	0	0	0	(0, 0, 0)	0	0	3 steps

F4 is the cleanest compiler-specific diagnostic: aggregate support and word bridges remain unchanged, but the Lie bridge and per-channel response record the broken data-flow observable. F5 shows the complementary path effect: single-snapshot mismatch is zero, but the pass-path wall record detects that the reference crosses the simplifycfg transition while the target remains fixed at the pre-simplifycfg snapshot.

Compiler IR SOF Report Alignment is a structural comparison of control-flow and dependency preservation under declared alignment. It does not replace semantic equivalence checking or a verified compiler proof.

6 Portability and Failure Boundaries

6.1 Portability Criterion

A target domain follows the same construction when the required SOF realization and alignment contract can be made explicit:

1. identify a domain-justified sectorization;
2. identify the observable family and its semantics;
3. construct or designate a reference report;
4. construct or observe a target report;
5. form the admissible comparison object $\mathfrak{C}_{\text{cmp}} = (\mathcal{R}^*, \widehat{\mathcal{R}}, \Phi; \Theta)$;
6. compute the induced $\Delta_{\text{audit}} = \text{Compare}(\mathfrak{C}_{\text{cmp}})$; and
7. record the comparison and its claim boundary.

6.2 Cross-Domain Synthesis

The four controlled domains isolate distinct roles for the same alignment grammar:

Domain	Sector origin	Observable family	Distinct alignment contribution
GridWorld	Cell sectors in a finite transition system	Directional action operators	Spatial support mismatch, obstacle-wall deformation, and word/Lie bridge-channel contrast
SIR	Compartment sectors	Infection/recovery rate operators	Rate extinction wall versus smooth response-order crossing
Traffic	Intersection-node sectors	Directed signal-phase operators	Phase aliasing, timing distortion, and rate-order trajectory mismatch
Compiler IR	Basic-block sectors	CFG and def-use observables	Dual observable-family alignment and path-only pass-pipeline wall

Their common comparison map is

$$\mathfrak{C}_{\text{cmp}} = (\mathcal{R}^*, \widehat{\mathcal{R}}, \Phi; \Theta) \mapsto \Delta_{\text{audit}} \mapsto \text{typed diagnostic coordinates.}$$

The domains differ in sector origin and observable semantics, but the comparison-object type remains the same. The controlled comparisons therefore establish portability of the alignment grammar without identifying the underlying failure mechanisms across domains.

6.3 Legitimate Transformation Controls

The failure-mode controls ask whether a target departs from a declared reference. The before/after controls ask a different question: what observable structure changes under a transformation already declared legitimate? Each control stores the full raw signature and a separate contract residual.

Transformation	Δ_{supp}	Δ_{brw}	Δ_{brl}	Δ_{dep}	$\Delta_{\text{frz}} = (R_1, W, L)$	Δ_{cns}	Δ_{ctrl}	Res.
Compiler O0-like $\rightarrow$ O2-like	4	8	6	8	(+4, +8, +8)	0	8	0
Traffic $\rho : 0.5 \rightarrow 2.0$	0	0	0	0	(0, 0, 0)	0	8	0
GridWorld obstacle (2, 2) $\rightarrow$ (0, 0)	12	26	10	102	(-4, -6, +232)	0	24	0

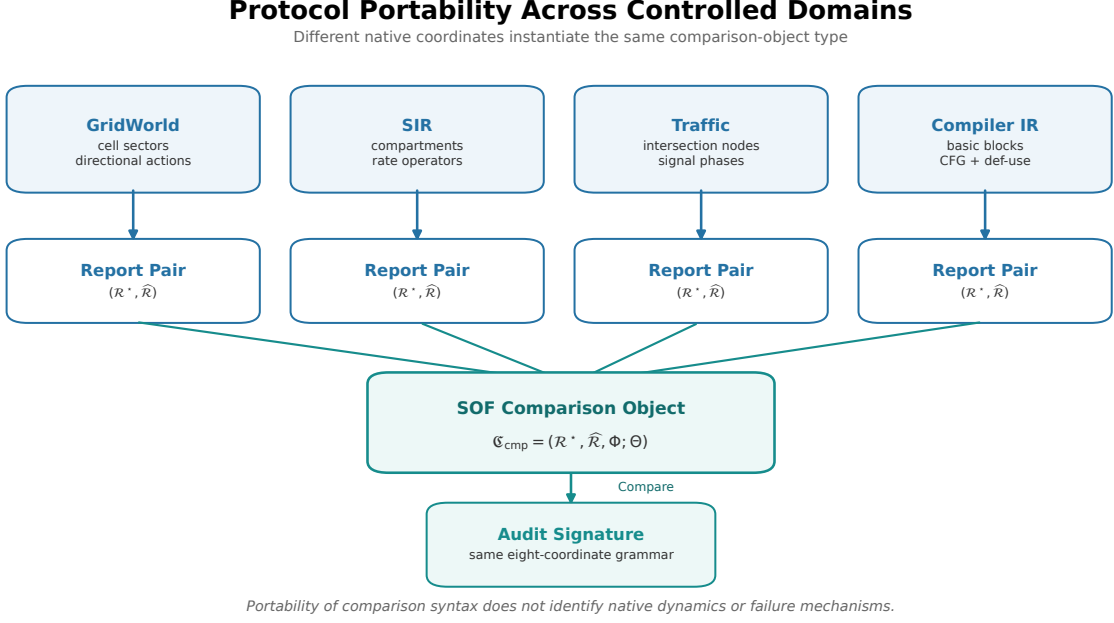


Figure 5. Protocol portability across controlled domains. GridWorld, SIR, Traffic, and Compiler IR use different sectors and observables, but each produces an explicit $\mathfrak{C}_{\text{cmp}}$ consumed by the same comparison map. The figure asserts portability of the comparison-object grammar, not equivalence of native dynamics.

The Compiler control uses a common five-block ambient realization with B3 marked as a retired target sector; its sector alignment is therefore explicit rather than silently treated as an unchanged active partition. Traffic preserves all binary topology while reversing the phase-response order. GridWorld changes support, bridges, depths, and frozen counts substantially, but every changed raw transition is incident to the declared old or new obstacle sector. In all three cases, nonzero Δ_{audit} means **changed**, while the zero contract residual means **no undeclared change was detected**.

6.4 Additional Validation Domain: Network Routing

Network Routing is included as an appendix validation domain rather than as a fifth member of the main synthesis table. It is useful because it isolates an ACL/policy-removal comparison role: the ACL acts as a constraint that removes edges from X_{external} , rather than an additive observable family like Compiler IR’s CFG/def-use pair.

The reference has four prefix sectors ($P0_{\text{local}}, P1_{\text{dmz}}, P2_{\text{internal}}, P3_{\text{external}}$) and two observables: X_{internal} for trusted-zone routes and X_{external} for gateway routes.

Failure	Δ_{supp}	Δ_{brw}	Δ_{brl}	Δ_{dep}	$\Delta_{\text{frz}} = (R_1, W, L)$	Δ_{cns}	Δ_{ctrl}	Δ_{wal}
F1 route aliasing	6	4	4	6	(+6, +6, +6)	3	12	–
F2 blocked prefix	2	0	0	2	(+2, 0, 0)	0	2	–
F3 forbidden route	0	2	4	0	(0, 0, 0)	1	2	–
F4 metric distortion	0	0	0	0	(0, 0, 0)	0	6	–
F5 ACL policy wall	0	0	0	0	(0, 0, 0)	0	0	4 steps

Two signatures are especially useful. F2 is a pure ACL edge-removal pattern: direct support changes, but word and Lie bridge channels do not. F3 has no direct-support mismatch, yet it produces word/Lie bridge mismatches plus a constraint violation. This makes the appendix domain complementary to Compiler IR rather than another traffic-shaped graph example.

6.5 Failure Boundaries

Not all domain-model pairs produce informative alignments. The failure modes documented in Paper XII’s Failure Modes and Applicability section (single sector, dense random, dim-1 sectors, commuting observables, sector-observable mismatch) apply to paired comparisons with additional constraints:

- **Alignment failure.** If Φ_{sec} or Φ_{obs} cannot be specified (incommensurable sectorizations or observable families), the paired comparison reduces to two independent single-system reports.
- **Coverage asymmetry.** If the reference covers state-space regions the target cannot represent (or vice versa), Δ_{frz} is dominated by coverage mismatch rather than structural error.
- **Rate-resolution floor.** Δ_{ctrl} requires per-control response constants above the tolerance floor. Models with near-zero rates produce Δ_{ctrl} indistinguishable from noise.
- **Threshold dependence.** The validation controls verify matrix-level sector permutation equivariance and conditional rescaling stability away from τ , but a multi-scale control changes structural support three times as τ crosses declared block magnitudes. Threshold stability is local, not universal.
- **Refinement boundary.** Coarse/fine sector comparison is not covered by the fixed-fiber proposition. It requires a pushforward law for structural signatures and remains open.
- **Cross-domain discrimination boundary.** Across the 25 failure-mode comparisons analyzed here, the normalized seven-coordinate structural vectors have same-label and different-label mean distances 1.0915 and 1.3991, a separation ratio of 1.2818. This is weak, domain-dominated separation. There is only one comparison per failure label within each domain, so no within-domain replication or clustering claim is made.

7 Conclusion

Paper XIII fixes the SOF comparison object $\mathfrak{C}_{\text{cmp}} = (\mathcal{R}^*, \widehat{\mathcal{R}}, \Phi; \Theta)$. Alignment supplies common retained coordinates, Θ fixes their comparison semantics, and $\Delta_{\text{audit}} = \text{Compare}(\mathfrak{C}_{\text{cmp}})$ is the induced output. On each fixed (Φ, Θ) fiber, the structural sub-signature carries a weighted pseudometric, while exact support inclusion and the single-generator X^2 witness separate word transport from Lie transport.

GridWorld, SIR, Traffic, Compiler IR, Network Routing, and the legitimate transformation controls demonstrate portability of the comparison grammar. They do not extend the pseudometric to the full signature or define comparison between fibers. Those extensions require normalization, symmetrization, and compatible pushforward laws.

Within the RIME program, Paper XII establishes reporting, Paper XIII establishes local aligned comparison geometry, and Paper XIV interprets nonzero comparison coordinates through context-indexed action semantics. This separation keeps difference, defect, and action as distinct objects.

8 Outlook

8.1 Alignment Regimes

The SOF comparison object applies to three reference-target regimes:

Regime	Reference	Target	Alignment status
A: controlled reference	explicit dynamics or structure	constructed variant or declared transformation	validated here
B: aligned latent model	environment or analytic reference	latent-state dynamics model	requires a justified latent-sector correspondence
C: black-box behavioral	grounded observations	API-, trajectory-, or video-level model	requires semantic probe alignment

These regimes classify the relation between the two reports; they are independent of the SOF Applicability Hierarchy, which classifies the quality of each source-to-report realization. Regimes B and C remain future applications. World-model proposals and executable-consequence benchmarks provide relevant contexts [9, 1, 13], but they do not supply evidence for the structural results proved here.

8.2 Comparison Tooling

A software implementation should preserve the mathematical order of operations: declare the report pair, record Φ and Θ , compute the signature, and only then apply domain-specific interpretation. Automatic alignment inference must remain distinguishable from a declared alignment, and trajectory comparison must retain parameter synchronization explicitly.

8.3 Open Problems

1. **Cross-fiber transport.** Define pushforward and transition laws that make structural signatures comparable across sector refinement and changing observable families.
2. **Alignment inference and stability.** Determine when Φ_{sec} and Φ_{obs} can be inferred, and how latent clustering or quantization uncertainty propagates to Δ_{audit} .

Alignment Regimes and Evidence Boundary

The comparison-object type is shared; the justification burden increases across regimes

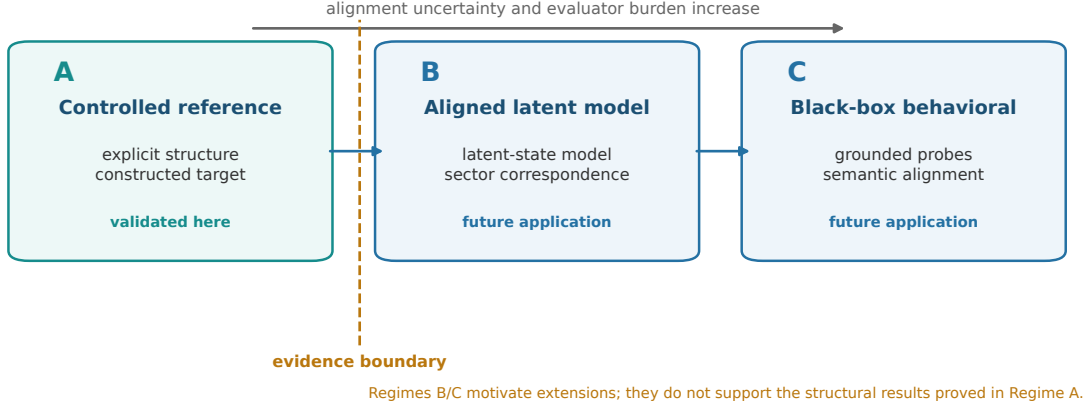


Figure 6. Alignment regimes and evidence boundary. Regime A supplies the controlled validation used in this paper. Regimes B and C share the comparison-object type but require increasingly strong latent-sector or semantic-probe justification; they remain application regimes rather than evidence for the fixed-fiber results.

3. **Full-signature geometry.** Extend the fixed-fiber pseudometric, if possible, to directional constraints, response coordinates, and synchronized wall records, with normalization across sector counts.
4. **Observable sufficiency.** Characterize minimal observable families that preserve diagnostically relevant distinctions and support causal attribution of response mismatches.
5. **Directed comparison costs.** Determine whether normalized channel strengths define canonical sector-pair costs without model-dependent weighting choices.

Appendix A Alignment Artifact and Schema Validation

The canonical machine-readable contract is `schemas/sofaudit/v1.0.schema.json`. A conforming `.sofaudit` record encodes an aligned comparison, not a second single-system SOF Report.

Contract component	Required content
Identity	contract version, artifact type, comparison-object type, audit identifier, and named system
Claim boundary	controlled claim status, nonempty qualification, regime, and failure modes
Report pair	reference and target report identifiers
Alignment	explicit sector and observable alignments
Comparison specification	normalization, thresholds, depth semantics, aggregation, and path synchronization when present
Audit signature	support, word bridge, Lie bridge, depth, frozen, constraint, response, and wall-record coordinates
Legitimate transformation	optional comparison role, transformation contract, and residual evaluation
Action boundary	no semantic interpretation, candidate action, risk, expected effect, or policy-selection fields

The fields `reference`, `target`, `alignment`, and `comparison_specification` serialize $\mathfrak{C}_{\text{cmp}}$; the field `signature` serializes Δ_{audit} . The contract does not use the single-report key `sofrs_version`, and a nonzero coordinate does not imply defect. Legitimate transformation controls retain their raw signature and record contract residuals separately. The contract rejects `decision`, `action_semantics`, `action_set`, and `selection`; these belong to the downstream `.sofaction` layer. Its top-level vocabulary is closed; the optional `domain` and `wall_note` fields are declared factual qualifications, not semantic or action outputs.

The validator entry point is:

```
python experiments/paper13/validate_sofaudit.py
```

Appendix B Experimental Reproduction

The following scripts and outputs constitute the reproducibility layer. Paths are relative to `experiments/paper13/`; generated reports and comparison records are stored in `results/`.

B.1 Controlled Domain Audits

- **GridWorld:** `gridworld_reference_sof.py` constructs the 25-sector reference and F1–F5 controls, producing six `.sofreport` and five `.sofaudit` records.
- **SIR:** `sir_compartment_sof.py` constructs the three-sector reference and F1–F5 controls, producing six `.sofreport` and five `.sofaudit` records.
- **Traffic:** `traffic_intersection_sof.py` constructs the four-sector reference and F1–F5 controls, producing six `.sofreport` and five `.sofaudit` records.
- **Compiler IR:** `compiler_ir_sof.py` constructs the five-sector reference and F1–F5 controls, producing six `.sofreport` and five `.sofaudit` records.
- **Network Routing:** `network_routing_sof.py` constructs the additional routing reference and F1–F5 controls, producing six `.sofreport` and five `.sofaudit` records.
- **Legitimate transformations:** `before_after_alignment.py` constructs three before/after controls, producing six `.sofreport` and three `.sofaudit` records.

B.2 Mathematical and Validation Controls

- `signature_metric.py` checks the fixed-fiber structural pseudometric and its stated boundary.
- `word_lie_controlled.py` checks the exact-support inequality, finite-threshold caveat, strict word-only witness, and GridWorld F4 record.
- `validation/stability_controlled.py` checks permutation equivariance, conditional rescaling, and threshold transitions.
- `validation/discrimination_controlled.py` checks coupled coordinates, legitimate transformations, and channel-specific controls.
- `validation/discrimination_analysis.py` provides the cross-domain boundary analysis over 25 comparisons; it does not support a within-domain replication claim.
- `validation/distance_profile.py` computes a declared-channel cost diagnostic that is excluded from the report metric.

- `validate_sofaudit.py` validates the canonical schema and all `.sofaudit` records.
- `regenerate_tables.py` regenerates `results/signature_tables.md`.

Appendix C Geometric Boundary and Outlook

The computational controls establish fixed-fiber and declared-threshold properties only. Sector permutations and conditional rescalings can be tested inside a declared fiber, while threshold crossings can change the discrete signature and changes of alignment data move the representation to a different comparison domain.

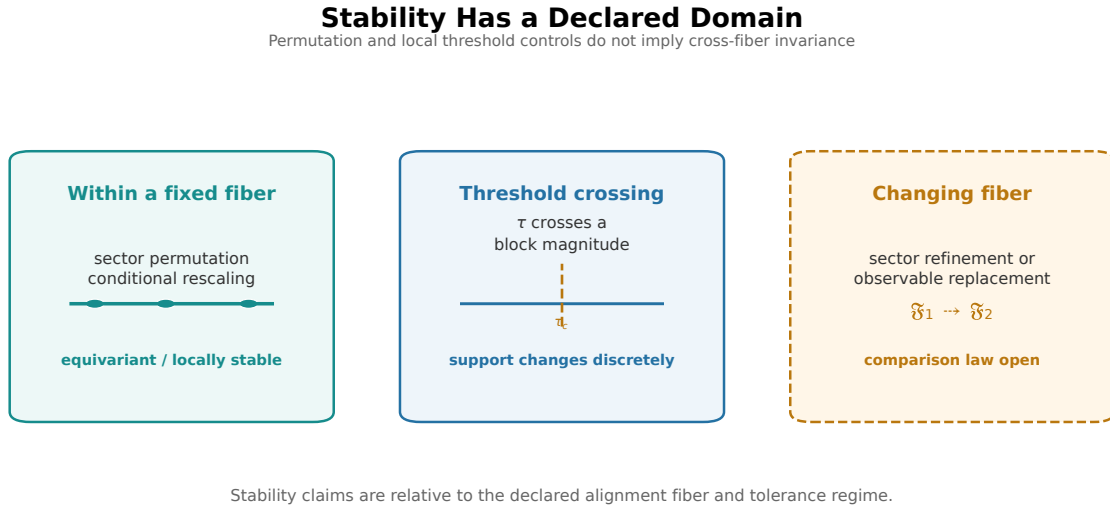


Figure 7. Stability boundary of the alignment protocol. Sector permutation and conditional rescaling are controlled within a fixed fiber, while threshold crossings can change discrete support. Sector refinement or observable replacement changes the comparison fiber and has no transition law in the present theory.

The resulting local domains do not yet form a global comparison geometry. Transition maps and pushforward laws must be defined before bundle, connection, transport, or curvature language can be promoted from outlook to theorem.

Comparison Geometry Beyond a Fixed Fiber

Paper XIII supplies local pseudometrics; transition and transport laws remain open

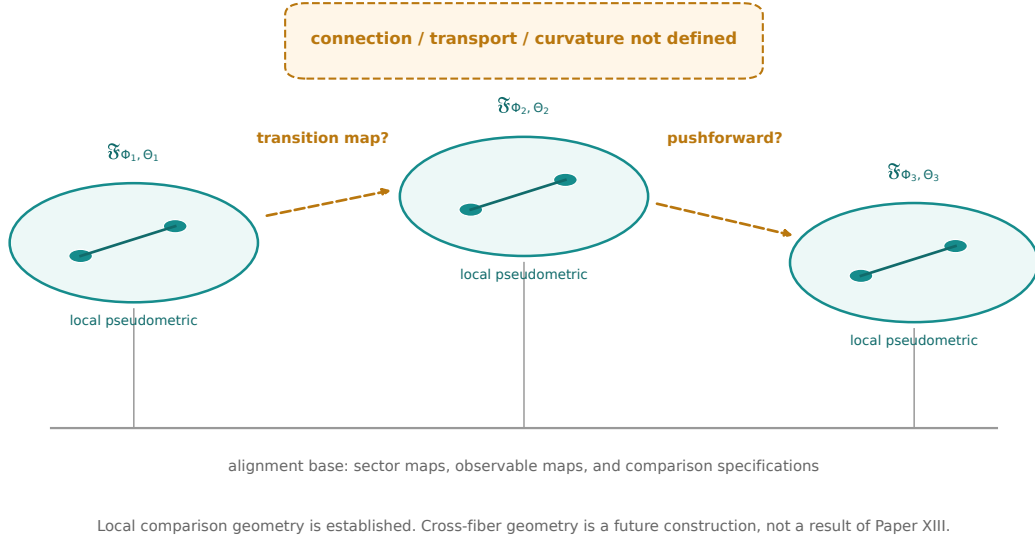


Figure 8. Comparison geometry beyond a fixed fiber. Each alignment fiber carries its own local structural pseudometric. Transition maps, pushforwards, connections, transport, and curvature between fibers are displayed as open constructions, not as results of Paper XIII.

Bibliographic Notes

Program lineage. Paper VIII supplies the SOF object layer; Paper IX supplies observable trajectories; Paper X supplies the Universal Observable Pipeline and Registry; Paper XI supplies wall records; and Paper XII supplies the single-report protocol consumed here [4, 3, 6, 2, 5].

Reporting and audit precedents. Model cards and end-to-end algorithmic audit frameworks provide precedents for structured records with explicit scope and failure boundaries [14, 16]. Paper XIII differs by treating sector and observable alignment as the mathematical prerequisite for a comparison signature.

Schema and ontology alignment. Database schema matching and ontology matching construct correspondences between fields, entities, and semantic relations [15, 7]. SOF Report Alignment has a different comparison unit: sector correspondences and observable correspondences must be declared together with threshold, depth, trajectory, and claim-status semantics before a comparison signature exists.

Representation matching. Orthogonal Procrustes alignment and modern representation-similarity methods compare coordinate representations or learned feature spaces [17, 11]. Paper XIII does not compare raw embeddings. It aligns declared observable structures and then compares their support, bridge, depth, frozen, response, constraint, and wall-record shadows.

Structured metrics. Hamming distance and graph edit distance provide standard precedents for discrete structural comparison [10, 8]. The fixed-fiber pseudometric here is more restricted than a general report distance: it is defined only on aligned structural representations under fixed (Φ, Θ) , and no metric is claimed on the full directional and path-dependent audit signature.

World-model deployment context. Latent-state world models and executable consequence benchmarks motivate the Regime B/C comparison problem [12, 9, 1, 13]. They are deployment contexts, not evidence for the fixed-fiber proposition or the word/Lie separation lemma.

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